

# Hot Days Turn Deadlier Where Guns Are Easy to Carry

Lessons from “Access to Guns in the Heat of the Moment: More Restrictive Gun Laws Mitigate the Effect of Temperature on Violence” by Jonathan Colmer & Jennifer L. Doleac in *The Review of Economics and Statistics*, 2026.

**Hot days lead to more gun homicides, but stricter concealed-carry laws blunt most of that effect.** On an average day, one degree of extra heat means more homicides nationwide — except in states that restrict who can carry a concealed gun, where that heat-driven increase lessens.

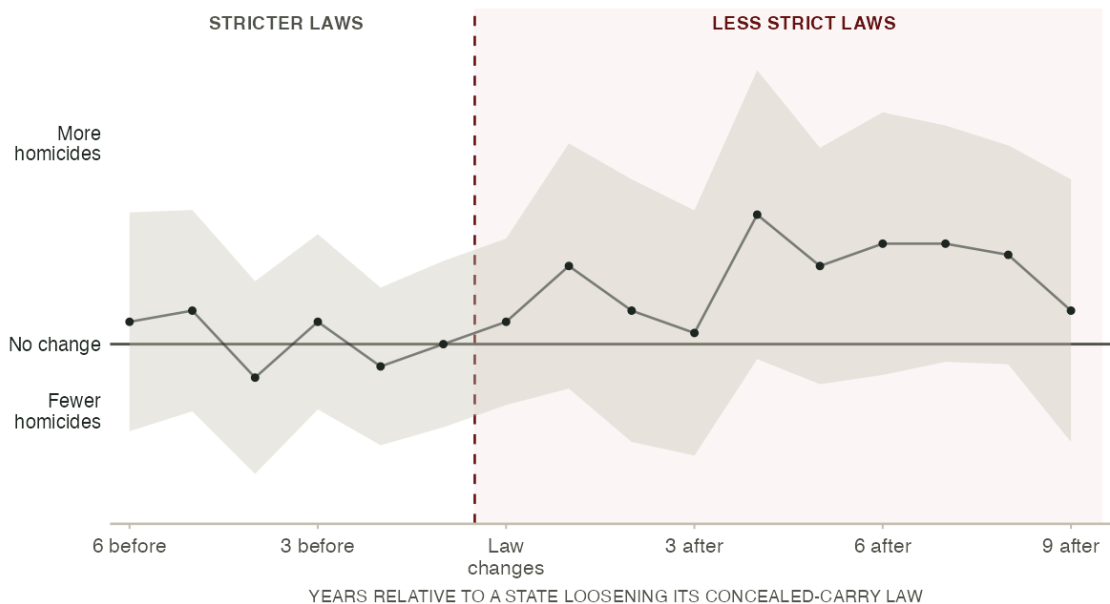
## BACKGROUND

Past research has established that high temperatures lead to increased aggression and violent behavior: hotter days are linked to more confrontations and violent crime. States also differ significantly in their **concealed-carry laws** — the rules governing who can carry a hidden handgun in public. Some states let nearly any adult carry one; others require a permit that officials can deny, or ban concealed carry altogether. Together, these two facts raise an important question: if heat makes people more likely to become violent, does having a gun close at hand change the results of these confrontations? To answer this question, researchers matched 25 years of daily police crime reports (1991–2016) with local weather records and each state’s concealed-carry laws, month by month. They focused on homicide because it is the crime type most consistently and reliably reported to police, across both place and time.

## WHAT THE STUDY FOUND

### HOW MUCH DEADLIER HOT DAYS BECOME AFTER A STATE LOOSENS ITS CONCEALED-CARRY LAW

Source: Colmer & Doleac (2026).



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*Note:* Each point tracks the gap in hot-day homicides between states that eventually loosen their concealed-carry law and states that keep theirs strict, year by year before and after the change (year 0). Before the change, the gap is flat — hot days are equally dangerous in both groups. Right after, it opens and stays open for years, which is what makes this look like the law’s effect rather than a coincidence.

In order to measure how stricter laws might change gun violence, researchers compared how much *extra* violence a hot day brings in strict-law places versus loose-law places. That comparison cancels out whatever else makes those states

different, as long as those other differences don't also change how a place responds to heat specifically. As a check, they also tracked the same states over time, comparing each state's own hot-day homicides before and after it changed its own law — shown in the figure above. Both approaches point to the same conclusion.

Nationally, each one-degree-Celsius increase in daily temperature is tied to roughly six percent more homicides than usual on an average day. In states with stricter concealed-carry laws, that heat-driven bump nearly disappears — both when comparing different states at the same time, and when tracking the same states before and after they changed their own laws. **Stricter concealed-carry laws blunt most of the effect of heat on homicides.**

This pattern is also concentrated specifically in gun homicides — not knives or other weapons, and not aggravated assaults broadly. That suggests strict laws aren't stopping hot-day arguments from happening in the first place; they're making the arguments that do happen less likely to turn deadly.

## TAKEAWAYS

### A GROWING RISK

Concealed-carry laws have loosened nationwide since the 1990s — from 36 states with strict laws in 1991 to just 10 by 2016 — even as climate change brings more hot days each year. Applying 2016's looser laws to 2016's population, instead of 1991's stricter mix, implies roughly 200 more homicides nationwide for every degree hotter (299 versus 97). **As gun laws loosen and summers heat up, the protection stricter laws once offered is being used less, right when it may matter most.**

### PUTTING A PRICE ON A HOT DAY

Using standard estimates of the cost of a homicide to society (about \$11.3 million each), the authors calculate that avoiding these heat-driven deaths nationwide would be worth roughly \$3.4 billion for every one-degree-Celsius rise in temperature — comparable to the estimated social benefit of hiring nearly 13,000 more police officers. The authors are clear this is a rough, illustrative calculation, not a precise cost-benefit accounting. **Even by conservative estimates, the stakes of heat-driven gun violence are large.**

### AGGRESSION, NOT MORE ENCOUNTERS

Why does heat make hot days more lethal where guns are easy to carry? It could be that people simply see more people when the weather is nice, or that heat makes people more aggressive once a conflict starts. The evidence favors the latter: the effect holds for homicides at home, not just in public; rainy days, which keep people inside, don't show the same pattern; and the effect is strongest in the hottest hours of the day. Additionally, there is no decrease in non-gun related homicides or aggravations. **While the heat makes people more aggressive, these conflicts get more lethal when guns are nearby.**

### WHAT THIS DOES (AND DOESN'T) SHOW

This study is about impulsive, heat-driven homicides. It doesn't speak to premeditated crime or gun violence more broadly. It also doesn't weigh the costs to gun owners of more restrictive laws, or say whether stricter concealed-carry laws should be adopted; that tradeoff is a policy judgment the authors leave open. **What the research does show is that temperature is a real, measurable driver of gun violence — and how much it matters depends on the laws already in place.**